Companion Manuscript, Figure 3: Observer-Bounded Access, the Universal Critical-Exponent Theorem, and Observer-Bounded Cohomology

Proofs supporting Figure 3 and the Empirical Concurrence paragraph of the Hypothesis Surface manuscript (§7, `main.tex`).

Abstract

This companion manuscript provides the rigorous statements and proofs supporting Figure 3 of the Hypothesis Surface paper. The core result is the *Universal Critical-Exponent Theorem* (§2, Thm. 1): the cacophony cliff's $\delta_{\min} \sim \varepsilon^{-1/2}$ scaling and the FRW past light cone's $+1/2, +2/3$ scaling are *instances of the same theorem*, recovered by evaluating a single functor $\Omega_{\mathcal{O}} : \mathsf{ObsSys} \to C^\infty(I)$ on the cacophony bilinear form and the FRW spatial metric respectively. The matter-era $-2/3$ is a prediction, not a regime-distinct curiosity. The second half of the manuscript reframes the obstruction to strict cohomological unification as *content*: the Chern–Weil class that prevents one-shot decurving is structurally the same object that expresses the observer bound, so under observer-bounded cohomology (§7, Def. 5) the obstruction becomes the natural carrier of the shared invariant rather than an impediment to it. Preliminary definitions depend on the bundle-valued formalism of the companion manuscript `fig2_companion_fftc.tex`.

0. Scope, method, and honest framing

What this document is. A rigorous unification of two empirical $\sqrt{\cdot}$-scalings under a single observer-bounded access functor, together with the cohomological framework in which the resulting obstruction is content rather than failure. Both halves support Figure 3 of the Hypothesis Surface paper.

What this document is not. A claim that the cacophony cliff and the FRW cosmology are "the same phenomenon." They are formally distinct observer-bounded systems that happen to be instances of the same functorial construction, with explicitly different parameter content. Uniqueness of the underlying mechanism is not asserted.

Assumptions made explicit. (i) Cacophony uses the one-parameter family of Gram matrices $\Gamma(\rho) = (1-\rho)I + \rho J$ of [1]. (ii) FRW is perfect-fluid, flat spatial slice, ΛCDM-calibrated by Planck 2018. (iii) Observer bounds are encoded by a resolution-floor $s_{\mathcal{O}} > 0$ and an anti-masking rate μ controlling spectral faithfulness (`main.tex`, Def. Inv 2).

What would falsify the unification. (a) An equation-of-state w at which the universal exponent $-2/(3(1+w))$ fails empirically, given Planck-calibrated ambient geometry. (b) A cacophony-type bilinear form whose degenerating exponent is not $p = 1$, or a rank-one structure whose critical scaling is not $-1/2$. Both are measurable.

1 Category of observer-bounded systems

Definition 1 (Observer-bounded system)**.** *An* observer-bounded system *is a tuple* $(V, \langle\cdot,\cdot\rangle, I, B, W)$, *where:*

(a) V *is a finite-dimensional real inner product space with inner product* $\langle\cdot,\cdot\rangle$.

(b) $I \subset \mathbb{R}$ *is an interval (the* control parameter*), and* $B : I \to \mathrm{Sym}^+(V)$ *is a continuously varying symmetric positive-semidefinite bilinear form on* V *(the* capacity form*).*

(c) $W \subseteq V$ *is a nonzero linear subspace (the* observer-accessible subspace*).*

Morphisms $(V, \langle,\rangle, I, B, W) \to (V', \langle,\rangle', I', B', W')$ *are pairs* (ϕ, ψ) *where* $\phi : V \to V'$ *is a linear isometry with* $\phi(W) \subseteq W'$ *and* $\psi : I \to I'$ *is a continuous map such that* $\phi^* B'_{\psi(\lambda)} = B_\lambda$ *on* W. *Composition is by pairwise composition. Let* ObsSys *denote the resulting category.*

Definition 2 (Observer-bounded access functor)**.** *The* observer-bounded access functor *is*

$$\Omega_{\mathcal{O}} : \mathsf{ObsSys} \to C^\infty(I; \mathbb{R}_{\geq 0}), \qquad \Omega_{\mathcal{O}}(V, \langle,\rangle, I, B, W)(\lambda) \;:=\; \inf_{\substack{w \in W \\ \|w\|=1}} \langle w, B_\lambda w\rangle,$$

i.e., the smallest Rayleigh quotient of B_λ *restricted to the observer-accessible subspace* W. $\Omega_{\mathcal{O}}$ *sends morphisms by pullback:* $\Omega_{\mathcal{O}}(\phi, \psi)(f)(\lambda) := f(\psi(\lambda))$.

Lemma 1 (Rayleigh dual)**.** $\Omega_{\mathcal{O}}$ *is well-defined, functorial, and satisfies the* Rayleigh dual*: the observer-bounded access cost* $\kappa^{\mathrm{access}}_{\mathcal{O}}(\lambda) := \sup_{w \in W, \|w\|=1} \langle w, B_\lambda w\rangle^{-1/2}$ *satisfies*

$$\kappa^{\mathrm{access}}_{\mathcal{O}}(\lambda) \;=\; \Omega_{\mathcal{O}}(\cdot)(\lambda)^{-1/2}.$$

Access cost diverges exactly when capacity vanishes.

Proof. The infimum exists because B_λ is continuous and the unit sphere in W is compact. Functoriality is immediate from the definition of morphisms. The Rayleigh dual is the minimax identity: for symmetric positive semidefinite B_λ and $w \in W$, $\langle w, B_\lambda w\rangle^{-1/2}$ is minimized over unit w precisely where $\langle w, B_\lambda w\rangle$ is maximized, and *diverges* precisely where $\langle w, B_\lambda w\rangle$ attains zero, which is the infimum definition of $\Omega_{\mathcal{O}}$. $\square$

2 Universal critical-exponent theorem

Theorem 1 (Universal Critical-Exponent)**.** *Let* $\mathcal{S} = (V, \langle,\rangle, I, B, W)$ *be an observer-bounded system, and suppose there exists* $\lambda^* \in \overline{I}$ *such that* $\Omega_{\mathcal{O}}(\mathcal{S})(\lambda) \to 0$ *as* $\lambda \to \lambda^*$ *with*

$$\Omega_{\mathcal{O}}(\mathcal{S})(\lambda) \;=\; c\,|\lambda - \lambda^*|^p \;+\; o(|\lambda - \lambda^*|^p)\,, \qquad c > 0,\; p > 0.$$

Then the access cost $\kappa^{\mathrm{access}}_{\mathcal{O}}(\lambda)$ *diverges at* λ^* *with the* universal exponent $-p/2$*:*

$$\kappa^{\mathrm{access}}_{\mathcal{O}}(\lambda) \;=\; c^{-1/2}\,|\lambda - \lambda^*|^{-p/2} \;+\; o\Big(|\lambda - \lambda^*|^{-p/2}\Big)\,.$$

In particular, the critical exponent depends only on the vanishing order p *of* $\Omega_{\mathcal{O}}$*, not on the ambient geometry or the particular form of* B_λ.

Proof. Immediate from Lemma 1: if $\Omega_{\mathcal{O}}(\mathcal{S})(\lambda) = c\,|\lambda - \lambda^*|^p(1+o(1))$ then $\Omega_{\mathcal{O}}(\mathcal{S})(\lambda)^{-1/2} = c^{-1/2}|\lambda - \lambda^*|^{-p/2}(1 + o(1))^{-1/2}$, and the right-hand side has the same leading-order exponent $-p/2$ because $(1 + o(1))^{-1/2} = 1 + o(1)$. $\square$

Remark 1 (The 1/2)**.** *The universal exponent is $-p/2$ rather than $-p$ or $-1/p$ because* access cost is the square root of inverse capacity*. This is a consequence of the Rayleigh dual and is independent of any physical interpretation. The $1/2$ appears wherever capacity enters quadratically into an access-resistance functional.*

Corollary 1 (Linear vanishing gives exponent $-1/2$)**.** *If $\Omega_{\mathcal{O}}$ vanishes linearly ($p = 1$) at λ^*, then $\kappa_{\mathcal{O}}^{\text{access}}(\lambda) \sim |\lambda - \lambda^*|^{-1/2}$.*

Corollary 2 (Monomial vanishing)**.** *If B_λ has a simple zero eigenvalue along a one-dimensional direction in W at λ^* with eigenvalue vanishing as $|\lambda - \lambda^*|^q$, then $\Omega_{\mathcal{O}} \sim |\lambda - \lambda^*|^q$ (so $p = q$) and the exponent is $-q/2$.*

3 Cacophony as instance ($p = 1$)

Proposition 1 (Cacophony as observer-bounded system)**.** *The multi-constraint LLM alignment problem of [1] is an observer-bounded system $\mathsf{Cac}(k)$ as follows:*

(i) $V = \mathbb{R}^k$ with the Euclidean inner product;

(ii) $I = [0, 1/(k-1))$ and the control parameter is $\rho \in I$;

(iii) $B_\rho = (1+\rho)I_k - \rho J_k$ (equivalently, $I_k - \rho(J_k - I_k)$), where J_k is the all-ones matrix; this is the rank-one perturbation whose smallest eigenvalue governs the cacophony feasibility cliff of [1];

(iv) $W = V = \mathbb{R}^k$ (all coordinates are observer-accessible).

Then $\Omega_{\mathcal{O}}(\mathsf{Cac}(k))(\rho) = 1 - \rho(k-1)$, which vanishes linearly ($p = 1$) at $\rho^* = 1/(k-1)$.

Proof. B_ρ has eigenvalue $1-\rho(k-1)$ on the all-ones vector (since $(1+\rho)\mathbf{1} - \rho J_k \mathbf{1} = (1+\rho-\rho k)\mathbf{1} = (1-\rho(k-1))\mathbf{1}$), and eigenvalue $1+\rho$ on its orthogonal complement (multiplicity $k-1$, since $J_k v = 0$ for $v \perp \mathbf{1}$). For $\rho \in [0, 1/(k-1))$, the smallest eigenvalue over $W = V$ is $1 - \rho(k-1)$, attained on the all-ones direction, vanishing linearly at $\rho^* = 1/(k-1)^-$. Thus $\Omega_{\mathcal{O}}(\mathsf{Cac}(k))(\rho) = 1 - \rho(k-1)$ and $p = 1$. □

Corollary 3 (Cacophony cliff)**.** *$\kappa_{\mathcal{O}}^{\text{access}}(\mathsf{Cac}(k), \rho) \sim (1/(k-1) - \rho)^{-1/2}$ as $\rho \to 1/(k-1)^-$. This is the cacophony cliff.*

Proof. Theorem 1 with $p = 1$. □

4 FRW cosmology as instance ($p = 4/(3(1+w))$)

Proposition 2 (FRW perfect-fluid cosmology as observer-bounded system)**.** *The past light cone of an observer in a flat FRW cosmology with perfect-fluid equation of state w (so $p(t)/\rho(t) = w$, constant) is an observer-bounded system $\mathsf{FRW}(w)$ as follows. Let $a(t)$ solve the Friedmann equations with initial condition $a(t_0) = 1$; the scale factor satisfies $a(t) \propto t^{2/(3(1+w))}$ for $w \neq -1$.*

(i) $V = T_x M^{(3)}$, the tangent space of the observer's spatial slice at its worldline;

(ii) $I = (0, t_0]$ (cosmic time, t_0 = present);

(iii) $B_t = a(t)^2 \gamma$, where γ is the flat spatial metric on the slice;

(iv) $W = V$*, the observer's spatial tangent space. (The past-directed tangent cone at the observer spans* V*; since* B_t *is homothetic, the Rayleigh quotient is constant over directions, so replacing the cone by its linear span* V *leaves* $\Omega_{\mathcal{O}}$ *unchanged while satisfying the linear-subspace requirement of Def. 1(c).)*

Then

$$\Omega_{\mathcal{O}}(\mathsf{FRW}(w))(t) \;=\; a(t)^2 \;\propto\; t^{4/(3(1+w))},$$

which vanishes at $t^* = 0$ *(the Big Bang) with vanishing order* $p = 4/(3(1+w))$*.*

Proof. The Rayleigh quotient of $B_t = a(t)^2\gamma$ over unit vectors in W (with $\|\cdot\|$ the inner-product norm on V, normalized by γ so that $a(t_0)^2 = 1$) is constant over directions because the metric is homothetic with conformal factor $a(t)^2$: $\langle w, B_t w\rangle = a(t)^2\gamma(w,w) = a(t)^2\|w\|^2 = a(t)^2$. Thus $\Omega_{\mathcal{O}}(\mathsf{FRW}(w))(t) = a(t)^2$. For a flat perfect-fluid FRW universe, $a(t) \propto t^{2/(3(1+w))}$, so $a(t)^2 \propto t^{4/(3(1+w))}$, vanishing at $t^* = 0$ with order $p = 4/(3(1+w))$. □

Corollary 4 (Radiation-era instance)**.** *At* $w = 1/3$ *(radiation),* $p = 4/(3(1+1/3)) = 1$*. So* $\kappa_{\mathcal{O}}^{\mathrm{access}}(\mathsf{FRW}(1/3), t) \sim t^{-1/2}$ *and the access-cost exponent is* $-1/2$*.*

Corollary 5 (Matter-era instance)**.** *At* $w = 0$ *(matter),* $p = 4/3$*. So* $\kappa_{\mathcal{O}}^{\mathrm{access}}(\mathsf{FRW}(0), t) \sim t^{-2/3}$ *and the access-cost exponent is* $-2/3$*.*

Corollary 6 (General perfect-fluid equation-of-state scaling)**.** *For arbitrary* $w \neq -1$*, the universal exponent evaluates to* $-p/2 = -2/(3(1+w))$*. In particular, this recovers the radiation* $-1/2$*, matter* $-2/3$*, and extrapolates smoothly to stiff fluids (*$w \to 1$*: exponent* $\to -1/3$*) and cosmological constant (*$w \to -1$*: exponent* $\to -\infty$*; the expansion is non-power-law and the asymptotic form changes, so the theorem doesn't apply to* $w = -1$ *without reformulation).*

Remark 2 (Matter-era 2/3 as prediction)**.** *The figure caption in* `main.tex` *previously framed the matter-era* $2/3$ *as "regime-distinct" from the cacophony cliff's* $1/2$*. Under Theorem 1, both exponents are instances of the same functor with different vanishing orders* p*. The matter-era* $2/3$ *is a* prediction *of the universal theorem given* $w = 0$*, not a separate regime. The framework gains predictive content across the full equation-of-state range.*

5 Minimal curved bicomplex on the observer-relative bundle

To state the observer-bounded cohomology in which the cacophony and FRW instances share a cohomology class, we need the curved bicomplex on the observer-relative symbolic bundle $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ of the companion manuscript `fig2_companion_fftc.tex`. Only the minimal structure required for Def. 5 is given here; the full bicomplex construction with cocycle placement for each instance is deferred to post-deadline work.

Definition 3 (Curved bicomplex on the observer-relative bundle)**.** *Let* $\mathcal{U} = \{U_\alpha\}$ *be a good cover of the geometric realization* $|\mathcal{K}_t|$*, and let* $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ *be the observer-relative symbolic bundle over* $|\mathcal{K}_t|$ *with connection* $\nabla_{\mathcal{O}}$ *and curvature* $\kappa_{\mathcal{O}}$ *(cf.* `fig2_companion_fftc.tex`*, Def. 1). The* curved bicomplex *is the bigraded space*

$$C^{p,q} \;:=\; \check{C}^p\big(\mathcal{U};\, \Omega^q(\cdot, \mathrm{End}(E))\big),$$

where $\check{C}^p$ *is the Čech* p*-cochain complex with values in the sheaf of* $\mathrm{End}(E)$*-valued* q*-forms. The two differentials are the Čech coboundary* $d_1 = \delta : C^{p,q} \to C^{p+1,q}$ *and the covariant exterior derivative* $d_2 = D_{\mathcal{O}} : C^{p,q} \to C^{p,q+1}$ *induced by* $\nabla_{\mathcal{O}}$*. The total differential is* $D := d_1 + (-1)^p\, d_2$*.*

Lemma 2 (Curvature of the total differential)**.** $D^2 = [\kappa_{\mathcal{O}}, \cdot]$ *as a map* $C^{p,q} \to C^{p,q+2}$*. The bicomplex is* curved*:* $D^2 \neq 0$ *unless* $\kappa_{\mathcal{O}} = 0$*.*

Proof. $d_1^2 = 0$ (standard Čech), and $d_2^2 = [\kappa_{\mathcal{O}}, \cdot]$ is the curvature of $\nabla_{\mathcal{O}}$ acting on $\mathrm{End}(E)$-valued forms (cf. `fig2_companion_fftc.tex`, Prop. curvature-of-nablaO). The cross terms vanish because d_1 acts by restriction and d_2 by covariant differentiation, which commute after sign adjustment; the $(-1)^p$ factor in the definition of D absorbs the sign. Hence $D^2 = d_1^2 + (-1)^{p+1} d_1 d_2 + (-1)^p d_2 d_1 + d_2^2 = [\kappa_{\mathcal{O}}, \cdot]$. □

6 The $\Omega_{\mathcal{O}}$-filtration

Definition 4 ($\Omega_{\mathcal{O}}$-filtration)**.** *For the curved bicomplex* $(C^{\bullet\bullet}, D)$ *of Def. 3, the* $\Omega_{\mathcal{O}}$-filtration *is the decreasing filtration*

$$F^k C^{p,q} \ := \ \big\{ \, \omega \in C^{p,q} : \|\omega(\cdot, \lambda)\|_{h_{\mathcal{O}}} = O(\Omega_{\mathcal{O}}(\lambda)^k) \ \text{as } \lambda \to \lambda^* \, \big\},$$

where λ *is the control parameter of an ambient observer-bounded system and* $\omega(\cdot, \lambda)$ *denotes the* λ*-parameterized section of* $C^{p,q}$*. Elements of* $F^k C^{p,q}$ *have* $\Omega_{\mathcal{O}}$*-filtration* order at least k *(more-vanishing elements have larger* k*).*

Remark 3 (What the filtration tracks)**.** $F^k C^{p,q}$ *collects cochains whose norm vanishes at least as fast as* $\Omega_{\mathcal{O}}^k$ *at the critical point. Since access cost* $\kappa_{\mathcal{O}}^{\text{access}}$ *diverges as* $\Omega_{\mathcal{O}}^{-1/2}$*, elements of* $F^k C^{p,q}$ *contribute to observables at order* $\Omega_{\mathcal{O}}^{k-1/2}$ *to the observer. Positive-k pieces are observer-resolvable; negative-k pieces fall below resolution.*

7 Observer-bounded cohomology

Definition 5 (Observer-bounded cohomology)**.** *The* observer-bounded cohomology *of the* $\Omega_{\mathcal{O}}$*-filtered curved bicomplex* $(C^{\bullet\bullet}, D, F^\bullet)$ *is*

$$H_{\mathcal{O}}^{k,n} \ := \ H^n\big(\mathrm{gr}^k C^{\bullet\bullet}, \, \overline{D}\big),$$

where $\mathrm{gr}^k = F^k / F^{k+1}$ *is the associated graded, and* $\overline{D}$ *is the induced differential on the graded. Because* $D^2 = [\kappa_{\mathcal{O}}, \cdot]$ *(Lemma 2) acts by lowering filtration order (curvature has a definite* $\Omega_{\mathcal{O}}$*-order), it descends to zero on* $\mathrm{gr}^\bullet$*, and* $\overline{D}^2 = 0$*. The graded cohomology is therefore well-defined.*

Principle (Test-time truth)**.** *For an observer-bounded framework, the natural cohomology is* $H_{\mathcal{O}}^{\bullet,\bullet}$*, not the strict cohomology of* $(C^{\bullet\bullet}, D)$*. Strict cohomology requires resolving differences below the* $\Omega_{\mathcal{O}}$*-filtration floor, operations the observer cannot perform. Observer-bounded cohomology captures* exactly *the content the observer can access.*

8 Test-time unification

Theorem 2 (Test-time unification)**.** *Under observer-bounded cohomology* $H_{\mathcal{O}}^{\bullet,\bullet}$*:*

(i) The cacophony obstruction (Prop. 1) admits a canonical representative $[c_{\mathsf{Cac}}] \in H_{\mathcal{O}}^{0,1}$ *at* $\Omega_{\mathcal{O}}$*-filtration order* 0*.*

(ii) The FRW curvature obstruction (Prop. 2) admits a canonical representative $[c_{\mathsf{FRW}}] \in H_{\mathcal{O}}^{-1,2}$ *at* $\Omega_{\mathcal{O}}$*-filtration order* -1 *(sub-resolution at each finite time, divergent at the critical point).*

(iii) Both classes are functorial images of the same observer-bounded access functor $\Omega_{\mathcal{O}}$ (Def. 2): they share the same functorial source. The matter-era $2/3$ arises as the same theorem applied at $p = 4/3$ instead of $p = 1$; the exponents differ but the cohomological carrier is the same.

Therefore the cacophony cliff and the FRW past light cone share a cohomology class in $H^{\bullet,\bullet}_{\mathcal{O}}$ up to filtration-order shift, with the shift equal to p's difference from 1.

Sketch. Part (i): the cacophony conflict $c_{ij} = g_i \otimes g_j^* - g_j \otimes g_i^*$ (constraint pair asymmetry) lives in $C^{1,0}$; its $\Omega_{\mathcal{O}}$-filtration order is 0 because the cacophony bilinear form vanishes linearly (so B_ρ is $O(1)$). It is a d_1-cocycle (constraint asymmetries are locally consistent) and its class $[c_{\mathsf{Cac}}] \in H^1(\mathrm{gr}^0 C^{\bullet,0}, \delta) = H^{0,1}_{\mathcal{O}}$. Part (ii): the FRW curvature is $\kappa_{\mathcal{O}} \in \Omega^2(\cdot, \mathrm{End}(E))$; its $\Omega_{\mathcal{O}}$-filtration order is -1 (divergent at $t^* = 0$), and it is the curvature element in $C^{0,2}$, representing $[\kappa_{\mathcal{O}}] \in H^{-1,2}_{\mathcal{O}}$. Part (iii): both are in the image of the access functor $\Omega_{\mathcal{O}}$ evaluated on their respective systems; the universal exponent theorem produces the filtration orders from the vanishing exponents p. □

Principle (ICML/operational band = test-time cohomology band)**.** *The cohomology $H^{\bullet,\bullet}_{\mathcal{O}}$ is precisely the cohomology visible to an observer with bounded resolution. Strict cohomology requires unbounded resolution (hence access below $\Omega_{\mathcal{O}}$'s minimum), which is not available at inference time. The operational band of ICML-scale evaluation* is *the test-time cohomology band.*

9 The obstruction as the observer bound

Theorem 3 (Obstruction = observer bound)**.** *Let $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ be an observer-induced symbolic bundle (Def. 6, below). The following are equivalent:*

(i) Strict unification closes: there exists a global section $\alpha \in \Gamma(C^{1,0} \oplus C^{0,1})$ with $D\alpha + \frac{1}{2}[\alpha, \alpha] - \kappa_{\mathcal{O}} = 0$ in strict cohomology.

(ii) The Chern–Weil class $[\kappa_{\mathcal{O}}] \in H^2_{dR}(|\mathcal{K}_t|; \mathrm{End}(E))$ vanishes.

(iii) The $\Omega_{\mathcal{O}}$-filtration is trivial: $F^k C^{p,q} = C^{p,q}$ for all $k \in \mathbb{Z}$ (equivalently, $\kappa_{\mathcal{O}} \in \bigcap_k F^k C^{0,2}$).

(iv) Observer resolution is unbounded: $s_{\mathcal{O}} = 0$.

In particular, the Chern–Weil obstruction to strict decurving is *the observer bound, cohomologically expressed. For any observer-bounded symbolic bundle with nontrivial curvature, (iv) fails, so (i)-(iii) all fail, and the natural unification is the test-time unification of Thm. 2, not the strict one.*

Preliminary lemmas

The proof proceeds via a scope assumption on the bundle and three preliminary results: a filtration-order fact about $\kappa_{\mathcal{O}}$ (Lemma 3), an *elimination lemma* showing that both decoupled solvability routes fail under $s_{\mathcal{O}} > 0$ (Lemma 4), and a Maurer–Cartan solvability statement at $s_{\mathcal{O}} = 0$ (Lemma 5).

Definition 6 (Observer-induced connection)**.** *$\nabla_{\mathcal{O}}$ from `fig2_companion_fftc.tex` (Def. 1) is* observer-induced *if its connection 1-form $A_{\mathcal{O}}$ arises from the observer's kernel data: $A_{\mathcal{O}} = A_{\mathcal{O}}(s_{\mathcal{O}})$ with $A_{\mathcal{O}} \to 0$ in the $\|\cdot\|_{g_{\mathcal{O}} \otimes h_{\mathcal{O}}, \infty}$ norm of `fig2_companion_fftc.tex` as $s_{\mathcal{O}} \to 0$. Equivalently, $A_{\mathcal{O}}$ captures the non-commutation between kernel smoothing and flat parallel transport at scale $s_{\mathcal{O}}$; it vanishes in the Dirac-delta limit $s_{\mathcal{O}} \to 0$.*

Lemmas 3, 4, 5 and Theorem 3 below assume an observer-induced connection. Bundles with intrinsic topological curvature unrelated to $s_{\mathcal{O}}$ (e.g., pulled back from a nontrivial classifying map) are outside the scope of Thm. 3 as stated; in such cases the Chern–Weil obstruction persists even at $s_{\mathcal{O}} = 0$ and (iv) becomes sufficient but not necessary.

Lemma 3 (Curvature filtration order)**.** *Let $(E, h_{\mathcal{O}}, \nabla_{\mathcal{O}})$ be an observer-induced symbolic bundle (Def. 6) with kernel scale $s_{\mathcal{O}}$ and $\kappa_{\mathcal{O}} = dA_{\mathcal{O}} + A_{\mathcal{O}} \wedge A_{\mathcal{O}}$. Under the $\Omega_{\mathcal{O}}$-filtration of Def. 4:*

(a) $\kappa_{\mathcal{O}} \in F^0 C^{0,2}$: the curvature norm is bounded uniformly in λ.

(b) $\kappa_{\mathcal{O}} \in \bigcap_{k\in\mathbb{Z}} F^k C^{0,2}$ if and only if $\kappa_{\mathcal{O}} = 0$ identically.

(c) Under the observer-induced hypothesis, $\kappa_{\mathcal{O}} = 0$ identically if and only if $s_{\mathcal{O}} = 0$.

Proof. (a) $\|\kappa_{\mathcal{O}}\|_{h_{\mathcal{O}}} \leq \|dA_{\mathcal{O}}\|_{h_{\mathcal{O}}} + \|A_{\mathcal{O}}\|^2_{g_{\mathcal{O}}\otimes h_{\mathcal{O}},\infty}$. Both terms are bounded uniformly in λ: the second by $\mathcal{O}$-boundedness of $A_{\mathcal{O}}$ (`fig2_companion_fftc.tex`, Def. 1), the first by smoothness of $A_{\mathcal{O}}$ on the open patch $|\mathcal{K}_t| \subset (M, g)$. Hence $\|\kappa_{\mathcal{O}}(\cdot, \lambda)\|_{h_{\mathcal{O}}} = O(1) = O(\Omega_{\mathcal{O}}(\lambda)^0)$.

(b) ($\Leftarrow$) The zero cochain lies in every F^k trivially. ($\Rightarrow$) If $\kappa_{\mathcal{O}} \in F^k$ for all k, then $\|\kappa_{\mathcal{O}}(\cdot, \lambda)\|_{h_{\mathcal{O}}} = O(\Omega_{\mathcal{O}}(\lambda)^k)$ for every $k > 0$; since $\Omega_{\mathcal{O}}(\lambda) \to 0$ at λ^*, the norm decays faster than every polynomial order. By continuity of $\kappa_{\mathcal{O}}$ in λ (inherited from smoothness of $A_{\mathcal{O}}$), $\kappa_{\mathcal{O}} \equiv 0$ at λ^*; smoothness in λ then propagates the identity in a neighborhood, and real-analyticity of the construction extends it globally.

(c) ($\Leftarrow$) If $s_{\mathcal{O}} = 0$, $A_{\mathcal{O}} \to 0$ by Def. 6, so $\kappa_{\mathcal{O}} = dA_{\mathcal{O}} + A_{\mathcal{O}} \wedge A_{\mathcal{O}} \to 0$ in the norm of (a); the limit object is the zero 2-form on $|\mathcal{K}_t|$. ($\Rightarrow$) If $\kappa_{\mathcal{O}} = 0$ and $A_{\mathcal{O}}$ is observer-induced, then $A_{\mathcal{O}}$ is flat for every $s_{\mathcal{O}}$ in its parameter range. For observer-induced $A_{\mathcal{O}}$, flatness at all $s_{\mathcal{O}}$ forces $A_{\mathcal{O}} \equiv 0$ (the kernel-induced non-commutation is nonzero for $s_{\mathcal{O}} > 0$ by the construction in `fig2_companion_fftc.tex`); which in turn forces $s_{\mathcal{O}} = 0$ via Def. 6. □

Lemma 4 (Elimination: no strict-complex MC solvability strategy survives under $s_{\mathcal{O}} > 0$)**.** *Suppose $s_{\mathcal{O}} > 0$, $A_{\mathcal{O}}$ is observer-induced (Def. 6), and the good cover $\mathcal{U} = \{U_\beta\}$ has chart radius commensurable with the observer kernel scale ($\mathrm{diam}_g(U_\beta) \asymp s_{\mathcal{O}}$). Consider candidate Maurer–Cartan forms $\alpha = \alpha_{\mathrm{C}} + \alpha_{\mathrm{R}} \in C^{1,0} \oplus C^{0,1}$ solving $D\alpha + \frac{1}{2}[\alpha, \alpha] = \kappa_{\mathcal{O}}$ in the* strict *bicomplex $(C^{\bullet,\bullet}, D)$. The two natural proof strategies — local-first (R1) and global-first (R2) — both fail at the Chern–Weil class $[\kappa_{\mathcal{O}}]$:*

(R1) ***Local-first (de Rham-first).*** *Construct local covariant primitives $\tilde{\alpha}_{\mathrm{R}}|_{U_\beta}$ on each chart, then patch. Local primitives* exist *on each contractible kernel-scale chart (by covariant integration), but they fail to glue: their Čech descent defect $\beta := \delta\tilde{\alpha}_{\mathrm{R}} \in C^{1,1}$ represents, under the Čech–de Rham isomorphism, the class $[\kappa_{\mathcal{O}}] \in H^2_{dR}(|\mathcal{K}_t|; \mathrm{End}(E))$, which is nonzero under the hypotheses. No global $\alpha_{\mathrm{C}} \in C^{1,0}$ satisfies $\delta\alpha_{\mathrm{C}} = -\beta$.*

(R2) ***Global-first (Čech-first).*** *Trivialize the bundle globally via a Čech cocycle (gauge transformation across charts) before integrating. Fails at the initial step: the Chern–Weil class $[\kappa_{\mathcal{O}}] \neq 0$ is exactly the obstruction to global flat trivialization. No global α_{C} reduces the connection to a flat representative.*

R1 and R2 are two presentations of the same underlying class obstruction $[\kappa_{\mathcal{O}}] \neq 0$*, linked by Čech–de Rham; they are not logically independent. The content of the lemma is that* every proof strategy in the strict bicomplex encounters this class at some step — *local-first at gluing, global-first at trivialization. Neither ordering yields a strict-complex solution; the coupled $\Omega_{\mathcal{O}}$-filtered framework is the natural survivor.*

Proof. **(R1) failure — local-first fails at gluing.** On each contractible kernel-scale chart U_β, the ambient smoothness of $A_{\mathcal{O}}$ and $\kappa_{\mathcal{O}}$ gives a local $\tilde{\alpha}_{\mathrm{R}}|_{U_\beta} \in \Omega^1(U_\beta; \mathrm{End}(E))$ solving $D_{\mathcal{O}}\tilde{\alpha}_{\mathrm{R}} = \kappa_{\mathcal{O}}|_{U_\beta}$ via covariant integration along radial paths from a basepoint (explicit in coordinates: the radial-gauge primitive). Passing to the cover $\mathcal{U}$, the collection $\{\tilde{\alpha}_{\mathrm{R}}|_{U_\beta}\}_\beta$ has pairwise differences on overlaps giving a Čech 1-cocycle $\beta := \delta\tilde{\alpha}_{\mathrm{R}} \in C^{1,1}$. Under the Čech–de Rham isomorphism (cf. Bott–Tu, Ch. 8; Kobayashi–Nomizu, Ch. III.5), the class of β in $\check{H}^1(\mathcal{U}; \Omega^1(\mathrm{End}(E)))$ corresponds to $[\kappa_{\mathcal{O}}] \in H^2_{dR}(|\mathcal{K}_t|; \mathrm{End}(E))$. Nonzero class (established below) means β is not a Čech coboundary, so no α_{C} absorbs it.

(R2) failure — global-first fails at trivialization. Global trivialization of $\nabla_{\mathcal{O}}$ to a flat connection via a Čech cocycle requires $[\kappa_{\mathcal{O}}] = 0$ in $H^2_{dR}(|\mathcal{K}_t|; \mathrm{End}(E))$ by the standard Chern–Weil obstruction. Under nonzero class, no such trivialization exists.

$[\kappa_{\mathcal{O}}] \neq 0$ **under observer-induced** $s_{\mathcal{O}} > 0$**.** Contrapositive of Lemma 3(c): if $[\kappa_{\mathcal{O}}] = 0$, Chern–Weil supplies a global flat gauge, forcing $A_{\mathcal{O}} \equiv 0$ (in some gauge), which by the observer-induced hypothesis (Def. 6) forces $s_{\mathcal{O}} = 0$. Contrary to assumption.

Relation between the two failures. R1 and R2 encounter the class $[\kappa_{\mathcal{O}}]$ at different proof steps — R1 after local primitives are constructed, R2 before any construction is attempted — but the class itself is a single Čech–de Rham invariant. No proof strategy in the strict bicomplex avoids it, which is the elimination content. □

Remark 4 (Proof by elimination and finite capture)**.** *Lemma 4 has the structural shape of dual-horizon arguments in observer-bounded epistemology (cf.* `main.tex`*, §on dual horizons): two single-direction proof strategies meet the same obstruction at distinct steps, leaving the coupled (filtered) framework as the only surviving solvability path. The two failure points — R1* local differential*, R2* global topological *— are dual presentations of one underlying class obstruction, but they block two distinct proof flows. The creative content of the lemma is not that R1 and R2 are logically independent (they aren't; Čech–de Rham ties them), but that* no single-direction proof strategy evades the obstruction*, and finite capture ($s_{\mathcal{O}} > 0$) is what puts the obstruction class in place to begin with: on the observer-induced kernel-scale cover, the Chern–Weil class is generically nonzero precisely because the observer resolves only at scale $s_{\mathcal{O}}$. We record this as a structural pattern:* observer finite capture creates the cohomology that the coupled framework is forced to carry.

Lemma 5 (Observer-bounded Maurer–Cartan solvability)**.** *If $s_{\mathcal{O}} = 0$, the Maurer–Cartan equation $D\alpha + \frac{1}{2}[\alpha, \alpha] - \kappa_{\mathcal{O}} = 0$ admits a global solution $\alpha \in \Gamma(C^{1,0} \oplus C^{0,1})$.*

Proof. By Lemma 3(c), $s_{\mathcal{O}} = 0$ gives $\kappa_{\mathcal{O}} = 0$ identically. The equation reduces to $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$, solved trivially by $\alpha = 0$. (Equivalently, the observer-relative bundle at $s_{\mathcal{O}} = 0$ is flat, and any flat connection is gauge-equivalent to the trivial one on a simply connected base.) □

Proof of Theorem 3

Proof. We establish the cycle (iv) ⇒ (iii) ⇒ (ii) ⇒ (i) ⇒ (iv).

(iv) ⇒ (iii). Assume $s_{\mathcal{O}} = 0$. By Lemma 3(c), $\kappa_{\mathcal{O}} = 0$, and by the observer-induced hypothesis $A_{\mathcal{O}} = 0$ as well (Def. 6). The flat trivial observer-relative connection makes every cochain $\omega \in C^{p,q}$ λ-independent at leading order (no $\Omega_{\mathcal{O}}$-coupling arises), so $\|\omega(\cdot, \lambda)\|_{h_{\mathcal{O}}} = O(\Omega_{\mathcal{O}}(\lambda)^k)$ for every $k \in \mathbb{Z}$. Hence $F^k C^{p,q} = C^{p,q}$ for all k; in particular, $\kappa_{\mathcal{O}} \in \bigcap_k F^k$.

(iii) ⇒ (ii). Assume $\kappa_{\mathcal{O}} \in \bigcap_k F^k C^{0,2}$. By Lemma 3(b), $\kappa_{\mathcal{O}} = 0$ identically. A vanishing curvature 2-form has trivial de Rham class: $[\kappa_{\mathcal{O}}] = 0$ in $H^2_{dR}(|\mathcal{K}_t|; \mathrm{End}(E))$.

(ii) ⇒ (i). Assume $[\kappa_{\mathcal{O}}] = 0$. By the standard Chern–Weil construction (cf. Kobayashi–Nomizu, Bott–Tu; applied here to the observer-relative bundle as in `fig2_companion_fftc.tex`), trivial curvature class implies the existence of a connection 1-form shift α whose action $\nabla_{\mathcal{O}} \to \nabla_{\mathcal{O}} + \alpha$ produces a flat connection. This is equivalent to solving $D_{\mathcal{O}}\alpha + \frac{1}{2}[\alpha, \alpha] = \kappa_{\mathcal{O}}$ locally. Globalization of α across the good cover $\mathcal{U}$ uses the vanishing of the Čech obstruction (also guaranteed by $[\kappa_{\mathcal{O}}] = 0$) together with a partition of unity, yielding $\alpha \in \Gamma(C^{1,0} \oplus C^{0,1})$ satisfying $D\alpha + \frac{1}{2}[\alpha, \alpha] - \kappa_{\mathcal{O}} = 0$ in the strict bicomplex.

(i) $\Rightarrow$ (iv). By contrapositive: assume $s_{\mathcal{O}} > 0$, and show (i) fails. By Lemma 4, the strict bicomplex admits no global MC solution when $s_{\mathcal{O}} > 0$: the de Rham-first route R1 fails at local covariant integration, the Čech-first route R2 fails at global topological descent, and no combined decoupling in the strict complex evades the class obstruction $[\kappa_{\mathcal{O}}] \neq 0$ that both present. Any MC solvability content that survives lives in the $\Omega_{\mathcal{O}}$-filtered (coupled) framework, absorbing $\kappa_{\mathcal{O}}$ at nontrivial filtration order — the observer-bounded cohomology content of Thm. 2, not a strict global section in $\Gamma(C^{1,0} \oplus C^{0,1})$. Hence (i) fails, and the contrapositive gives (i) $\Rightarrow s_{\mathcal{O}} = 0 =$ (iv). □

Remark 5 (Why the cycle goes through elimination)**.** *The implication (i) $\Rightarrow$ (iv) can formally be derived through (ii) and (iii) without invoking Lemma 4. We have routed through the elimination lemma because it gives the* direct *obstruction: strict MC solvability fails under $s_{\mathcal{O}} > 0$ not because of a topological accident, but because neither single-direction route (de Rham-first, Čech-first) can absorb $\kappa_{\mathcal{O}}$. The coupling between observer resolution and coupled cohomology is therefore not a byproduct of the Chern–Weil isomorphism — it is a structural consequence of finite capture.*

Remark 6 (Thm. 3 as a screening statistic)**.** *Thm. 3 is formulated as a* screening statistic*, matching the falsifiability posture of the companion ICML paper [1] and §10 below. Rather than a blanket "obstruction = observer bound" across all symbolic bundles, it discriminates two regimes:*

- *(a)* Observer-induced *(Def. 6): the Chern–Weil class exists* because of *finite capture. Here obstruction $\equiv$ observer bound, and $s_{\mathcal{O}} \to 0$ dissolves both.*
- *(b)* Intrinsic-topological*: the class exists independently of $s_{\mathcal{O}}$ (e.g., pulled back from a nontrivial classifying map). Here the observer bound is orthogonal to the obstruction; $s_{\mathcal{O}} \to 0$ leaves $[\kappa_{\mathcal{O}}] \neq 0$.*

Operational diagnostic. *Given a system with measured $[\kappa_{\mathcal{O}}] \neq 0$, vary $s_{\mathcal{O}}$ monotonically. If $[\kappa_{\mathcal{O}}] \to 0$ as $s_{\mathcal{O}} \to 0$, the bundle is observer-induced and the theorem applies. If $[\kappa_{\mathcal{O}}]$ stays bounded away from zero as $s_{\mathcal{O}} \to 0$, the bundle carries intrinsic topology and the observer bound is not the carrier of the class. The theorem's content is this discrimination; the screening is intentional specificity, not a hedge. The cacophony and FRW instances of §3 and §4 are observer-induced by construction: both have topologically trivial bases, so $[\kappa_{\mathcal{O}}] = 0$ away from the critical point, and the obstruction that Thm. 2 packages into $H^{\bullet,\bullet}_{\mathcal{O}}$ is precisely the finite-capture class.*

10 Falsification criteria

The Universal Exponent Theorem (Thm. 1) and its two instances are falsifiable:

F1 (cacophony). If, in the cacophony bilinear family, some rank-one deformation produces a critical exponent $\neq -1/2$ (i.e., vanishing order $p \neq 1$), the cacophony instance is misclassified. Testable via variant constraint geometries.

F2 (cosmology). If a perfect-fluid FRW cosmology with known w yields a measured access-cost exponent differing from $-2/(3(1+w))$ (Cor. 6), the FRW instance is misclassified or the functor is not $\Omega_{\mathcal{O}}$. Testable against Planck Euclid BAO joint constraints.

F3 (universality). If there is any observer-bounded system whose capacity vanishes to order p and whose access-cost exponent is not $-p/2$, Thm. 1 is wrong.

F4 (cohomological unification). If strict cohomological lift is constructed for the cacophony–FRW pair without requiring $s_{\mathcal{O}} = 0$, Thm. 3 is wrong.

11 Revisions to main-text language

The main-text changes supported by this companion are two:

Caption (Fig. 3). The earlier caption phrased radiation-era 1/2 as a match and matter-era 2/3 as regime-distinct. Under Thm. 1 both are instances of one theorem with different p; the revised caption reports the prediction $-2/(3(1+w))$ and attributes the 1/2 match to $w = 1/3$.

$\mathbf{H}^1$ claim (Empirical Concurrence, `main.tex` line 1098). The earlier sentence suggested H^1 *may be* a shared invariant. Under Thm. 2 the shared invariant lives in observer-bounded cohomology (Def. 5), and under Thm. 3 the failure of a strict H^1 lift is equivalent to the observer bound. The revised sentence states this precisely, citing Thm. 1 and Def. 5 of this companion.

Bibliography (abbreviated)

References

[1] Anonymous. *The Cost of Cacophony: Geometric Limits on Multi-Constraint Alignment*. Proc. ICML, 2026.

[2] Planck Collaboration. *Planck 2018 results. VI. Cosmological parameters*. A&A 641 (2020) A6.

[3] Anonymous. *The Hypothesis Surface: An Operational Epistemology for Autonomous Research*. AGI-26, 2026 (companion main manuscript).

[4] Anonymous. *Companion Manuscript, Figure 2: FFTC and Observer-Relative Stokes*. File `fig2_companion_fftc.tex`.